

Week 5- Data Governance- Part 2

Dr. Smitha K G

Senior Lecturer,

College of Computing and Data Science,
Nanyang technological University

Responsible AI frameworks

Responsible AI ensures that AI systems are not only technically effective, but also socially acceptable, legally compliant, and ethically sound.

Responsible AI in Banking (Credit Approval)

- AI provides a risk score, not a final decision
- Applicants receive clear explanations (e.g., income stability, credit history)
- Bias testing across demographic groups
- Human officer reviews borderline cases
- Decisions and models are audited

Responsible AI in Education (Student Risk Prediction)

- AI flags students for additional academic support
- Outputs are private and advisory
- Students are informed
- No automated penalties
- Ethical review approval obtained

Data Governance is about how data is managed, controlled, and protected

Responsible AI is about how AI systems behave and affect people

Core Ethical Considerations in Responsible AI

Human-Centricity

- AI should support, not replace, human judgment
- Humans must be able to intervene

Privacy

- Controlled, consent-aware data use
- Prevent data leakage, inference attacks, re-identification

Security & Robustness

- AI should behave reliably
- Risks such as drift, misuse, and adversarial attacks must be addressed

Accountability

- Clear responsibility for AI outcomes
- Human oversight must exist
- “The algorithm decided” is **never an acceptable answer**

Ethics: Fairness & Non-Discrimination

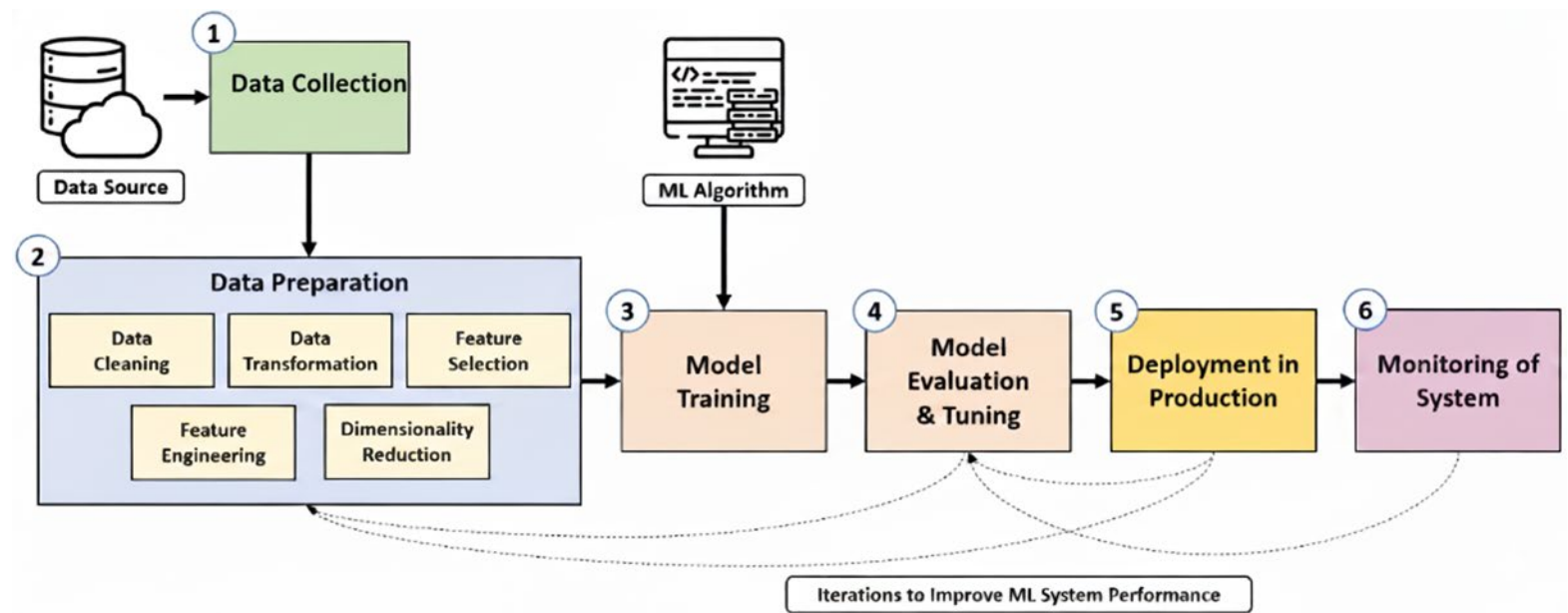
- AI should not systematically disadvantage groups
- Avoid direct and proxy discrimination

Psychology: Transparency & Explainability

- Users should know when AI is used
- Decisions should be explainable at an appropriate level
- **No need to** reveal source code but meaningful explanation of *why* a decision was made



Responsible AI Across the Data science pipeline

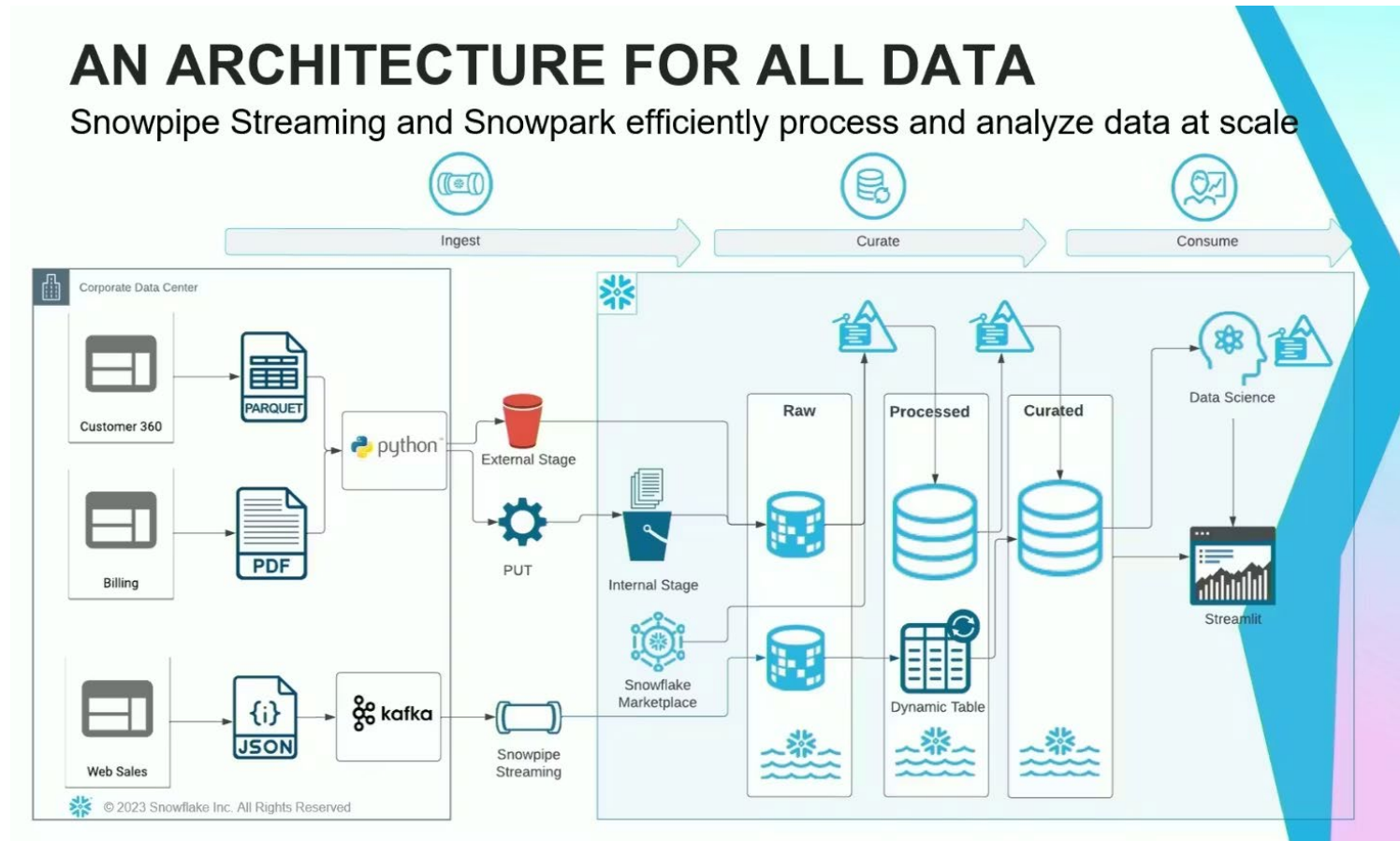


Data Science Pipeline	Responsible AI Question
Problem definition	Should this decision be automated?
Data collection	Is consent sufficient and fair?
Feature engineering	Are we encoding bias?
Model training	Are we testing subgroup performance?
Deployment	Is human oversight present?
Monitoring	Are harms emerging over time?

What is AI Data storage?

AI data storage is specialized infrastructure built to feed data-hungry AI systems efficiently.

- Designed for AI & ML workloads
- Handles massive data volumes
- Enables fast access for training & inference
- AI training repeatedly reads large datasets
- Inference often needs low-latency access
 - Ex: Fraud detection systems must access transaction data in milliseconds
 - Medical imaging AI must load scans instantly



[AI Data Storage: Benefits, Challenges & Best Practices](#)

AI Data storage- Details

Designed to handle very large volumes of data

- Example: A retail platform stores years of customer clickstream data for recommendation systems.

High-performance and scalable storage for fast access

- Example: A self-driving car company retrieves terabytes of sensor data daily for model training.

Supports efficient training and inference

- Example: A healthcare AI system quickly loads medical images to assist doctors during diagnosis.

Uses data reduction techniques to optimize cost and efficiency

- Deduplication: Removing repeated medical images in a hospital archive
- Compression: Compressing satellite imagery used in climate modelling
- Tiering: Frequently used training data on SSDs, archived data on cheaper storage

AI Data storage- Details

Strong emphasis on data security

- Example: A financial institution encrypts customer data used for credit-risk models.

Access control to restrict who can use AI data

- Example: Only authorized data scientists can access student performance data in an educational AI system.

Integration with AI/ML frameworks

- Example: Data stored in a cloud data lake (the example taken) is directly accessed by TensorFlow or PyTorch pipelines.

Optimizes cost while maintaining performance

- Example: A startup uses cloud-based AI storage to scale training only when needed, avoiding large upfront costs.

Improves overall reliability of AI systems

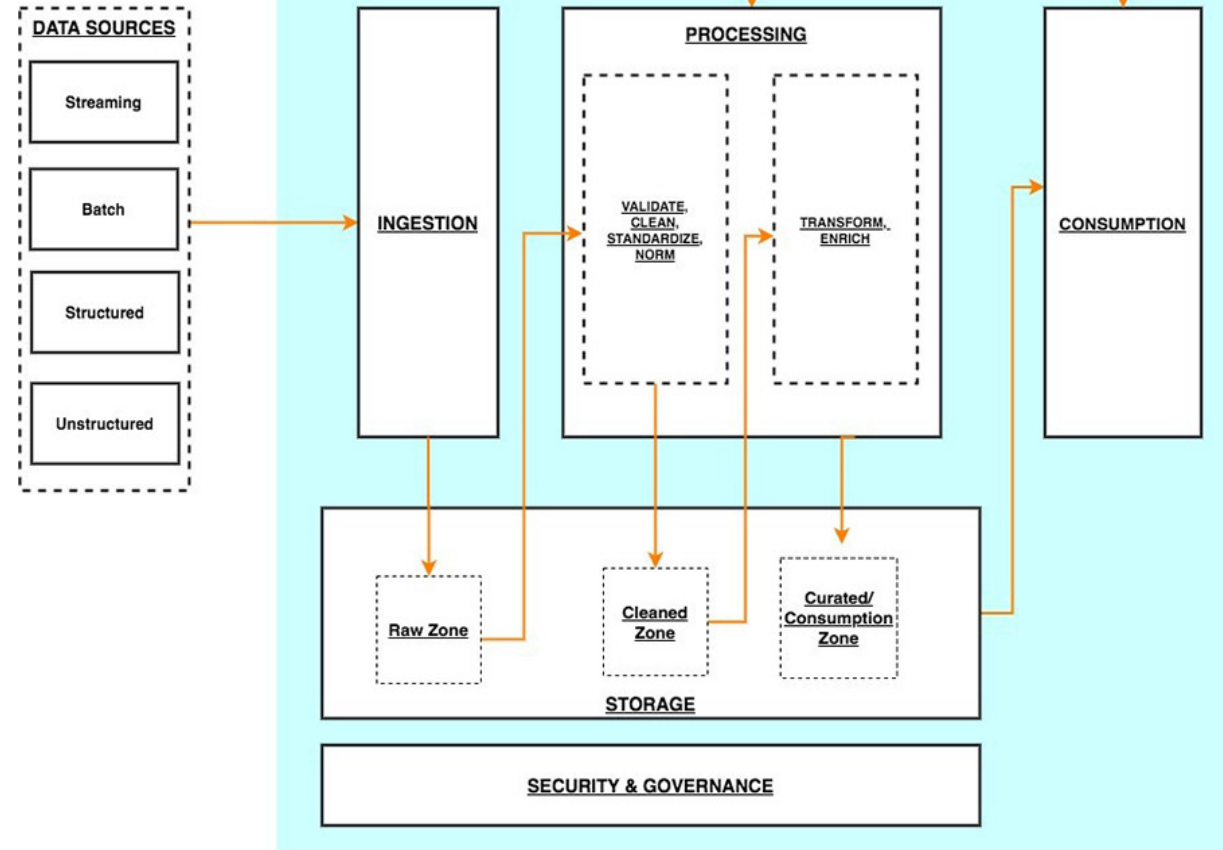
- Example: An e-commerce company ensures recommendation models remain responsive even during peak sales events.

AI Data storage- Scalability

- AI data grows faster than traditional business data.
- Supports scaling from GBs to PBs
- Used across industries: retail, finance, healthcare
- Cloud and hybrid storage are common

Issues to address

- Cost & Efficiency Optimization (Not all AI data needs premium storage all the time.)
 - Deduplication → removes repeated data
 - Compression → reduces storage footprint
 - Tiering → hot / warm / cold data
- Security and Data protection
 - Encryption at rest & in transit
 - Role-based access control
 - Protection against leakage & misuse



AWS serverless technologies

Heart of AI Infrastructure

Modern AI is driven by massive parallel processing using GPUs — and storage exists to keep GPUs busy.

- GPUs are expensive and powerful
- Idle GPUs = wasted money
- Storage must feed data at GPU speed

AWS: EC2 P4d / P5 GPU instances rely on high-throughput storage to avoid GPU starvation

Azure : ND-series GPU VMs require fast data feeds from Azure Blob / Azure NetApp Files

Low-latency flash storage is critical for AI training and inference.

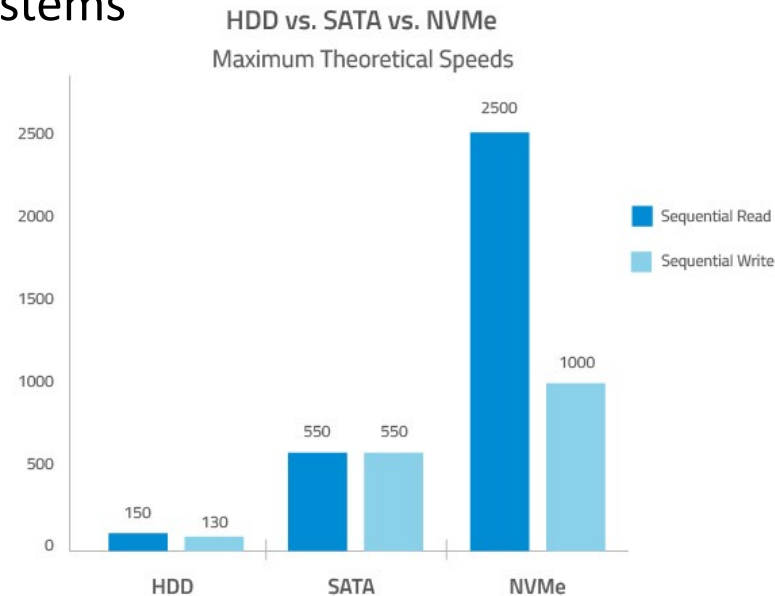
- AI workloads need low-latency I/O
- Flash (SSD / NVMe) is preferred over HDD
- Training loops repeatedly read the same data

AWS FSx for Lustre: NVMe-backed, optimized for GPU training

Azure NetApp Files: Ultra-low latency for AI pipelines

Low-latency flash storage

- NVMe (Non-Volatile Memory Express) is a high-speed storage protocol designed specifically for flash memory (SSDs) that connects directly to the CPU over PCIe lanes.
- AI is data-hungry and latency-sensitive
- NVMe feeds GPUs fast enough
- Massive parallelism (64K queues × 64K commands) where SATA only has less than 32 commands in a queue
- Energy & scale efficiency is good .
- Storage delay = GPU idle = wasted energy + money → NVMe: finishes I/O faster, lowers idle timescales better in multi-GPU systems



Storage	Typical latency
HDD(slow)	~5–10 ms
SATA SSD	~100 μs
NVMe SSD	10–20 μs

AI Storage Capacity Is Massive. AI datasets quickly scale from hundreds of terabytes to petabytes.

- Training data + augmented data + outputs
- Intermediate checkpoints stored repeatedly
- Model artifacts grow over time

AWS S3 (Singapore region) → PB-scale AI data lakes

Azure Data Lake Storage Gen2

AI storage must support different I/O patterns, not just raw speed.

- Different frameworks stress storage differently:
- PyTorch → many small files
- TensorFlow → fewer, larger files
- Storage must handle: High IOPS
- High throughput
- Mixed file sizes

AWS FSx for Lustre optimized for small-file access (PyTorch)

Azure Blob + Azure ML Datastores optimized for TensorFlow pipelines

Secure data storage

Secure data storage refers to the technical and organizational controls used to protect AI data throughout its lifecycle—from collection and training to deployment and long-term retention—while ensuring compliance with data governance principles.

Why secure storage is critical?

Data is reused repeatedly

Data often contains personal or sensitive information

Models can leak data indirectly

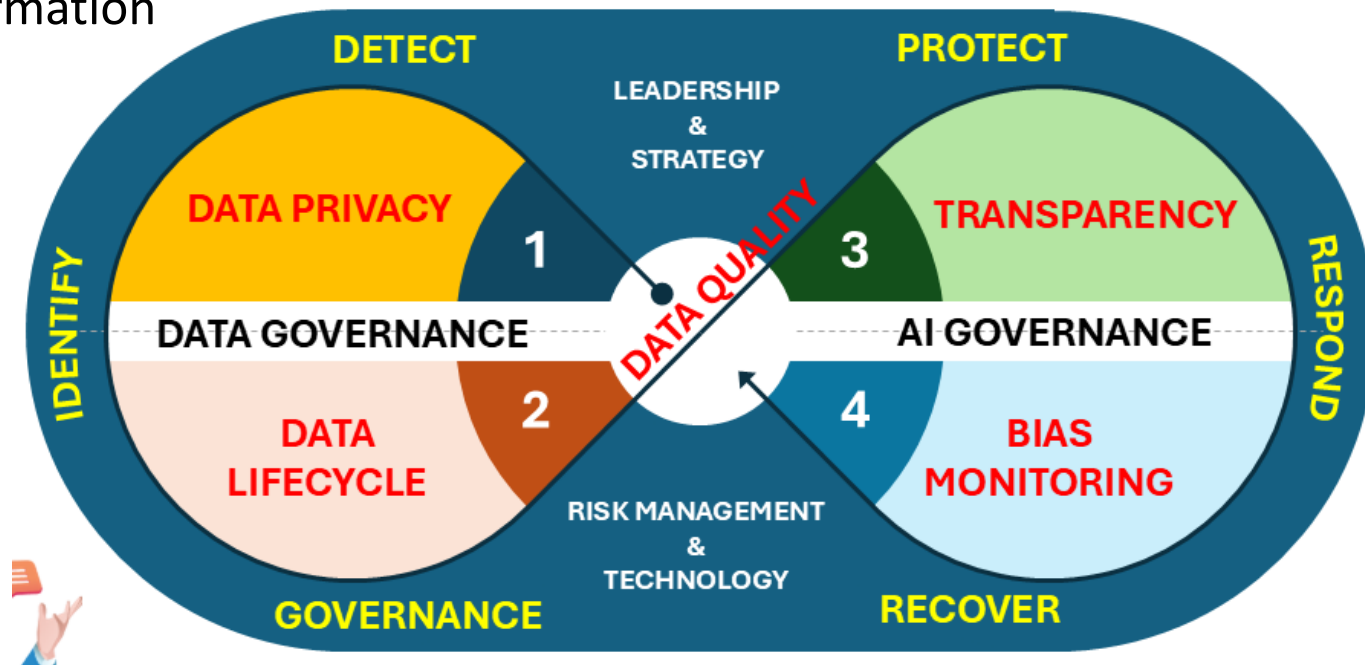
Secure storage enables:

Privacy protection

Regulatory compliance (e.g., PDPA)

Accountability and auditability

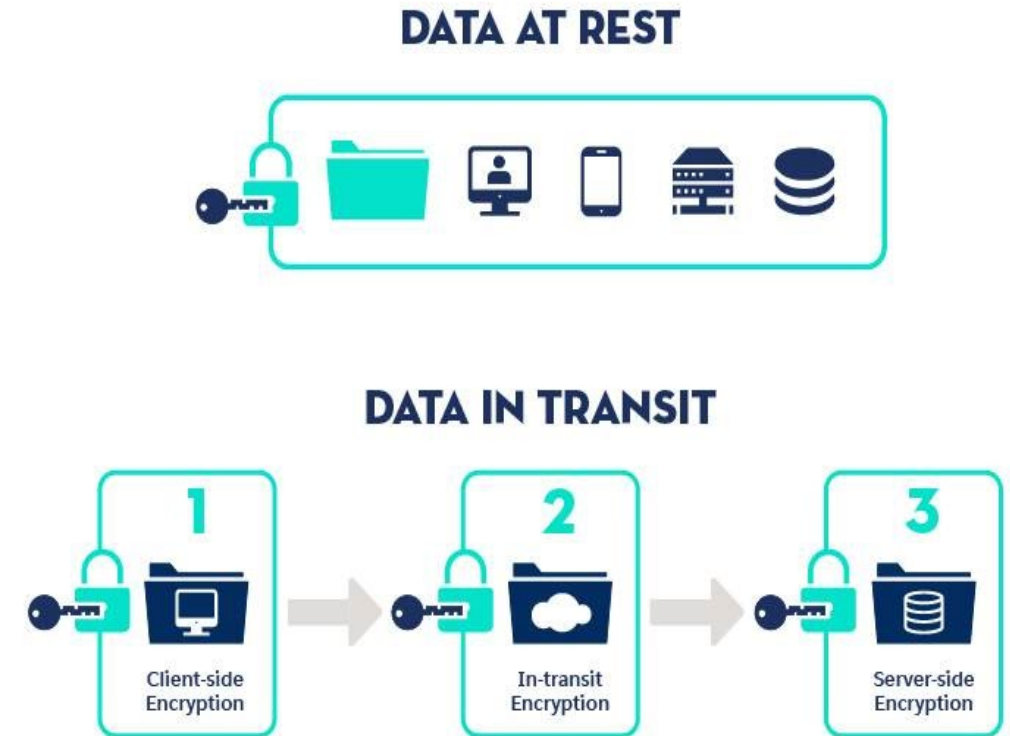
Trust in AI systems



Secure data storage

Zero Trust Architecture (ZTA)

- Core idea: Never trust, always verify — even inside the system.
- Key principles:
 - Identity-first security
 - Continuous verification
 - Least privilege access
- Data ingestion → validation → training → deployment (Each step authenticated separately)
 - Encryption (at rest & in transit)
 - At rest → protects stored datasets
 - In transit → protects data flowing to GPUs or cloud services



Secure AI Platform – Key Cornerstones

Infrastructure – The Foundation

- Provides the base on which AI models and applications run
- Includes secure compute, networking, and storage
- Comparable to the physical structure and access control of a storage facility

Data – The Protected Fuel

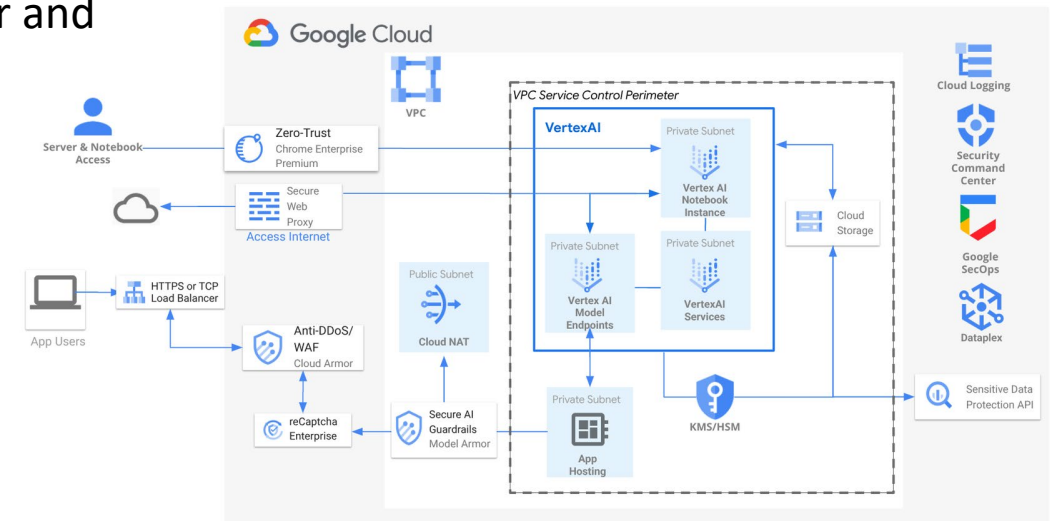
- Data is the core resource that powers AI systems
- Must be protected from unauthorized access, modification, and theft
- Data ensures AI model integrity, privacy compliance and customer and user trust

Security – The Shield

- Protects the entire AI ecosystem from threats
- Detects, prevents, and responds to attacks
- Focuses on: Reducing attack surfaces, maintaining confidentiality, integrity, and availability

Responsible AI (RAI) – The Ethical Compass

- Ensures AI systems are used ethically and intended purpose
- Builds long-term trust
- Prevents misuse or harmful outcomes



[Mastering secure AI on Google Cloud: A practical guide for enterprises | Google Cloud Blog](#)

Core Security Pillars for secure AI Data Storage

Access Control (Who can access data?)

- Only authorized users and systems can access data
- Role-based access (RBAC)
 - Prevents unauthorized model training
 - Prevents misuse of sensitive datasets

Audit Logs & Monitoring

- Every data access is logged
- Continuous monitoring for unusual behavior
 - Enables accountability
 - Supports compliance audits and investigations
- **Example:** Tracking who accessed a healthcare dataset and when during model training

Encryption (How data is protected)

- Data is encrypted: At rest (stored), on transit (moving between storage and GPUs), secure key management
 - Protects against data breaches
 - Required under PDPA's protection obligation
- **Example:** Medical images encrypted before being used to train diagnostic AI models

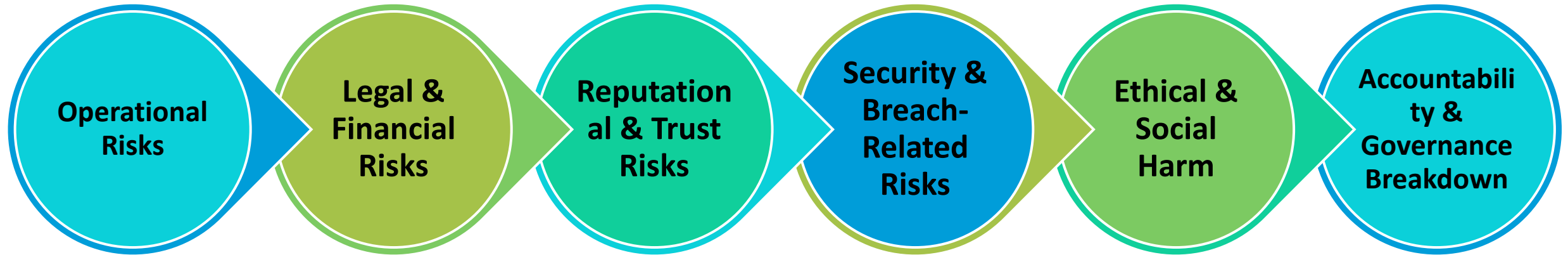
Data Isolation & Segmentation

- Separate environments for: Training, testing and production
- Sensitive data isolated from general datasets
 - Reduces blast radius if a breach occurs
 - Prevents accidental data leakage across projects
- **Example:** Research datasets stored separately from operational student records

Secure Retention & Deletion

- Clear data retention periods
- Secure deletion when data is no longer needed
- Models may retain learned information from deleted data
- May require model retraining or decommissioning
- **Example:** Student survey data deleted after research completion and results publication

Risks of Regulatory Non-Compliance



- Regulatory fines
- Enforcement actions
- Mandatory suspension of AI system

- Models must be retrained
- Data pipelines must be redesigned
- AI systems taken offline

- Loss of public trust
- Loss of customers or users
- Media scrutiny

- Data breach
- Model inversion or inference attacks
- Regulatory penalties + public disclosure

- Discriminatory outcomes
- Psychological harm
- Societal backlash

- No clear owner for failures
- Blame shifted to “the algorithm”
- Weak internal controls exposed

Risks of Regulatory Non-Compliance

Risk Type	What Goes Wrong	Real-World Impact
Legal	PDPA violations	Fines, shutdowns
Operational	Model invalidation	Retraining costs
Reputational	Public backlash	Loss of trust
Ethical	Discrimination	Social harm
Security	Data breaches	Legal + reputational damage
Governance	No accountability	Systemic failure

Advanced Topics in Data Governance and Responsible AI

AI Model Governance & Model Lifecycle Management

Governance isn't a checkbox at the end; it's the nervous system of the entire model lifecycle

Model versioning- How do you track a model that learns in real-time?

Approval workflows before deployment

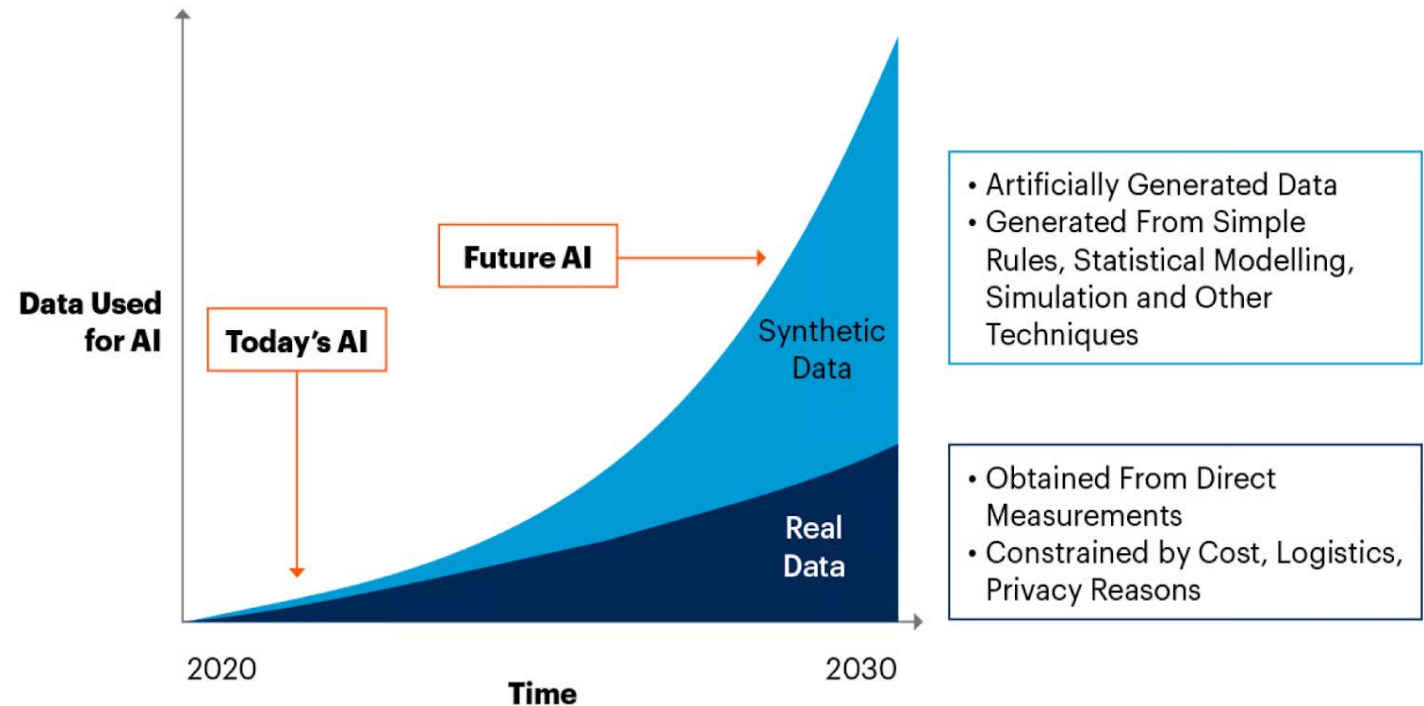
Model retirement and rollback- The "Kill Switch" strategy.

Governance of retraining- Data may be compliant, but model behavior may drift into non-compliance

Synthetic data generation

- Synthetic data is artificially generated data that mimics the structure and statistical patterns of real-world data
- It does not correspond to real individuals or entities
- Created algorithmically rather than collected from real-world events
- Used as a substitute or supplement for real data
- Enables software testing without using real user data
- Supports AI model training and data augmentation
- Useful in financial modeling where real data is confidential
- Widely used in healthcare research when patient data is sensitive
- Helps overcome data scarcity and privacy constraints.

By 2030, Synthetic Data Will Completely Overshadow Real Data in AI Models



[20 Generative AI Tools For Creating Synthetic Data](#)

1. Know the Purpose

- Clearly define **why synthetic data is needed** and where it is preferable to real data
- Useful when real data is: Sensitive (healthcare, finance, education), Scarce (rare events, fraud, edge cases) and Restricted by privacy, consent, or IP laws
- Enables: Safe data sharing, software testing without production data and data augmentation for underrepresented groups

2. Preparation Is Critical

- Synthetic data quality depends on real data quality
- Original datasets must be: Cleaned (remove errors, duplicates, completed (handle missing values) and include edge cases and outliers to reflect real-world variability

3. Diversify Data Sources

- Synthetic data can inherit biases from original datasets
- Mitigate bias by: using multiple data sources, including diverse demographics and regions
- Retrieval-augmented generation can improve accuracy and context

4. Beware of Model Collapse

- Repeated training on synthetic data alone can degrade model quality
- Always **ground synthetic data generation in real data**
- Use domain taxonomies or constraints to guide generation
- Synthetic data should *support* learning, not replace reality entirely.

5. Choose Appropriate Synthesis Techniques

- **Statistical methods:** suitable for simple, well-understood distributions
- **Generative Adversarial Networks (GANs):** used for images, vision tasks
- **Variational Autoencoders (VAEs) :** generate variations of existing data
- **Transformer models (e.g., GPT):** Generate synthetic text or tabular data
- Hybrid approaches are common in enterprises

6. Validate Synthetic Data Quality

- **Fidelity (Similarity to Real Data):** statistical similarity (distribution, variance, correlations), visual checks (histograms, plots)
- **Utility (Effectiveness for ML):** Train models on synthetic data, compare performance with real-data training using: Accuracy, Precision, Recall, F1 score, Image metrics (FID, Inception Score)
- **Human-in-the-Loop:** Manual inspection of random samples remains essential

7. Keep Privacy Central

- Synthetic data usually contains **no direct PII**, but risks still exist
- Must comply with privacy laws (e.g., GDPR, HIPAA, PDPA)
- Apply safeguards: Masking and anonymization, differential privacy, secure access controls

8. Document, Monitor, and Refine

- Maintain documentation of: Data preparation, generation techniques, privacy controls, validation results
- Enables: Transparency, auditability, reproducibility
- Continuously monitor downstream model performance and refine workflows

Differential Privacy

- **Differential Privacy** is a mathematical technique that **adds controlled noise** to data or query results so that **no single individual's data can be identified**, even if attackers have background knowledge.
- Widely adopted in: Government statistics, large tech platforms, healthcare and finance
- Helps comply with privacy laws (GDPR, PDPA, HIPAA)
- **Challenges**
 - More noise → better privacy, lower accuracy
 - Less noise → better utility, higher re-identification risk
- **Example**
 - A hospital releases synthetic patient statistics using differential privacy
 - If noise is poorly calibrated: Rare disease patterns may still be inferred, Population-level traits (e.g., genetic prevalence) can leak

Differential privacy protects individuals, but population patterns can still be revealed.

Federated Learning

- **Federated Learning** trains AI models **across multiple decentralized devices or institutions**, where data stays local and only model updates are shared
- The model moves, not the data, thus reducing raw data sharing
- Useful when data is: sensitive, legally restricted, distributed across organizations
- Attractive for healthcare, finance, and mobile AI
- **Challenges**
 - Model updates can still leak information
 - Requires: Secure aggregation, Strong authentication and robust monitoring
 - Harder to audit and debug than centralized training
- **Example**
 - Multiple hospitals jointly train a diagnostic model
 - Patient data never leaves each hospital
 - If updates are not properly protected, attackers may infer site-specific or population patterns

Federated learning reduces data movement, not governance responsibility.